# On the Algebraic Restoration of the Pure-Time Argument of Probability

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## Abstract

Pure time stood the very target for Hamilton's programme. This paper restores — in the literal sense of *al-jabr*, the reunion of separated parts that gives algebra its name — the argument of probability that Hamilton sketched in his 1844 paper on the calculus of probabilities, lifted to the pure-time coordinate system of the 1834 essay *Algebra as the Science of Pure Time*. The restoration takes the form of a single ordinary-least-squares regression on the pure-time basis $(\mathbf{y}_t, \Delta\mathbf{y}_t)$, algebraically equivalent to the autoregressive VAR(2) but with the error-correction matrix and the cointegration rank emerging directly from its OLS estimate. The paper is the practical face of an integrated programme whose theoretical foundations are developed in two companion papers — the first on non-abelian characteristic functions on Spin(3) (Companion paper A), the second on geometric invariants of the embedded trajectory (Companion paper B) — and whose philological framing is given in the synthesis manuscript of the GSD programme. Working in the algebra $\mathrm{Cl}(3,0)$ of the kinematic embedding $\mathbf{q}(t) = (y_t, \Delta y_t, \Delta^2 y_t)$ that those papers construct, we report:

1. **Identifiability invariant** (*$\gamma$-Theorem*). The rank of the empirical embedding matrix $Q \in \mathbb{R}^{(T-2)\times 3}$ is a discrete invariant of the underlying process matching the geometric Takens dimension $d^\dagger$ of [Researches in GSD, Representation]. In particular, $\operatorname{rank} Q < 3$ for purely sinusoidal, polynomial, or fixed-point dynamics; $\operatorname{rank} Q = 3$ for generic stochastic processes, sums of $\geq 2$ harmonics, and chaotic systems.

2. **Deterministic exactness** (*Battery D*). For deterministic data drawn from any model $y_t = f_{\theta_0}(t)$, the graded action loss vanishes exactly at $\theta_0$ and the resulting estimator recovers $\theta_0$ to numerical precision modulo the local optimiser; no statistical noise is necessary or available.

3. **Noise-robustness gain on rank-3 dynamics** (*Battery N*). Under additive Gaussian noise of scale $\sigma$, the grade-3 (pseudoscalar) component of the action delivers parameter estimates with systematically smaller variance than the level-1 (pointwise) match for chaotic-logistic-type processes; for sinusoidal (rank-2) DGPs the grade-3 component is degenerate and contributes only noise.

4. **Multivariate generalisation and emergent cointegration rank.** The forward equation lifts naturally to vector processes $\mathbf{y}_{t+1} = A\mathbf{y}_t + B\Delta\mathbf{y}_t + \boldsymbol{\varepsilon}_{t+1}$, equivalent to VAR(2) via $\Phi_1 = A+B$, $\Phi_2 = -B$. This re-parametrisation yields a key diagnostic by-product: the cointegration rank $r$ is read off as the effective rank of $\hat{\Pi} := \hat{A} + \hat{B} - I$ via SVD, with no prior specification. On six classical macro/financial cointegration test pairs (FRED yields, equity pairs, Argentina FX), the emergent rank recovers the textbook result (yield curve $r = d - 1 = 2$, stock pairs $r = 1$) and remains informative on borderline cases where the Engle–Granger ADF residual test fails.

5. **Multivariate quadratic extension for chaotic dynamical systems.** On the Lorenz '63, Rössler, and coupled-logistic systems, the same pure-time basis augmented with quadratic interaction terms fits the dynamics to numerical precision (in-sample RMSE $\sim 10^{-13}$ on noise-free Lorenz, $\sim 10^{-15}$ on coupled logistic).

The estimator is the natural re-parametrisation of OLS in the pure-time coordinate system: classical OLS-AR($p$) and VAR($p$) are algebraically equivalent to PT-univ and PT-MV respectively in the linear regression sense, and forecasts coincide to floating-point precision in well-conditioned regimes. The substantive gain of the pure-time parametrisation is interpretive and diagnostic: the coefficients of the forward equation map directly onto the cointegration matrix and the kinematic differential operator, exposing structure that the Cartesian-lag basis hides behind constraints imposed externally (Johansen's rank specification, regime-specific dispatch between OLS, Phillips–Perron, and Park-style estimators).

# 1 Introduction

## 1.1 The persistent regime problem

A practitioner facing a single univariate series $\{y_t\}$ must choose, before any estimation can begin, an assumption about its regime: stationary AR($p$), integrated of order 1 or 2, explosive, trending, structurally broken. Each assumption selects a different estimator: ordinary least squares for stationary AR; augmented Dickey-Fuller and unit-root tests for integration; Phillips-Perron and Park-style estimators for explosive regression. The same is true in the multivariate setting, where the chain fragments further: VAR($p$) for stationary panels, the Engle-Granger two-step or the Johansen procedure for cointegrated I(1) panels, explicit specification of the cointegration rank in advance.

This paper offers a single estimator that is well defined and $\sqrt{T}$-consistent under every regime listed above without modification, in both the univariate and the multivariate setting, and that returns as a by-product of unconstrained ordinary least squares the cointegration rank that Johansen's procedure usually asks the user to specify.

## 1.2 The pure-time perspective

The estimator is not new in the algebraic sense: it is OLS in a specific basis of regressors, the *kinematic embedding* $\mathbf{q}(t) = (y_t, \Delta y_t, \Delta^2 y_t)$. We recall this embedding from the Representation-Theorem manuscript of the GSD programme [Researches in GSD, Representation], where its sufficiency at dimension $d = 3$ was established as a representation theorem. The novelty of this paper is the observation, not formalised in earlier work, that two specific regressors taken from this embedding — $(y_t, \Delta y_t)$ in the univariate case and $(\mathbf{y}_t, \Delta \mathbf{y}_t)$ in the multivariate case — share the property that, for every regime in the list above, they have asymptotically distinct orders of integration. They are therefore well-conditioned where the classical Cartesian regressors $(y_{t-1}, y_{t-2})$ become asymptotically collinear (explosive, near-unit-root) or share an integration order (cointegrated panel).

This conditioning property has two faces. The first is the *transparency theorem* (Theorem 6.3): a single OLS regression is consistent across regimes. The second is the *emergent cointegration rank* (Theorem 3.1 and Definition 7.1): the rank of the OLS estimate of the matrix $\Pi = A + B - I$ recovers the cointegration rank of the system without prior specification.

The basis $(y_t, \Delta y_t)$ is, in the language of the Hamilton 1834 essay *Algebra as the Science of Pure Time* [Hamilton(1834)], an ordered pair of *steps in pure time* of orders 0 and 1. The closing remark of that essay announces that "a Theory of Triplets and Sets of Moments, Steps, and Numbers" would extend its treatment of *couples* — which is the substance of Hamilton's quaternions, foreshadowed nearly a decade before their formal discovery on 16 October 1843 at Brougham Bridge. The kinematic embedding is the modern realisation of Hamilton's announced programme; the present estimator is its statistical face.

## 1.3 Hamiltonian programme of pure time

This paper is the estimation arm of an integrated programme that reads classical time-series analysis through Hamilton's pure-time algebra. The companion paper A [Vázquez Broquá, A] establishes the non-abelian characteristic function of a probability law in $\mathrm{Cl}(3,0)$, generalising Cramér's abelian Fourier representation. The companion paper B [Vázquez Broquá, B] establishes the geometric invariants of the embedded trajectory, hodograph and Egregium curvature. The MSDS work [Vázquez Broquá, MSDS] establishes the moment-tensor reading. The synthesis manuscript

[Researches in GSD, Synthesis] provides the historical and philological framing. The present paper completes the picture on the side of *identification*: how to estimate the parameters of a process whose state lives in $\mathrm{Cl}(3,0)$.

The recurring observation across these works is that the classical-statistics canon emerges as the *abelian projection* ($j = 0$ in the SU(2) representation theory of the companion paper A) of a richer construct. Cramér's characteristic function is the $j = 0$ projection of the wedge-CF; the Pearson moments are the $j = 0$ projection of the moment tensor; OLS is the $w_0 = 1$, $w_2 = w_3 = 0$ specialisation of the graded action functional (Definition 2.4). Each of these projections preserves the abelian content of the parent construct exactly. The non-abelian projections, when they are non-trivial, contribute information that the abelian one does not.

Finding non-trivial uses of the non-abelian content of $\mathrm{Cl}(3,0)$ turns out to be subtler than it appears. The first part of this paper documents the $\gamma$-theorem and the deterministic and near-deterministic regimes (Sections 3 to 5) where higher-grade projections of the kinematic embedding genuinely contribute. The second part of the paper restricts to the abelian content of the embedding only — the linear regression on the $(y_t, \Delta y_t)$ basis — and exhibits the *transparency* property described above (Section 6 onward). The two parts are complementary: the higher grades carry information, but only for processes whose embedding is full rank, and the transparency property already exhausts the basic statistical identification problem at the abelian level.

### 1.4 The three schools of the programme

In the closing pages of *Algebra as the Science of Pure Time*, Hamilton distinguishes three schools by which the study of algebra may be pursued:

> "The Practical person seeks a Rule which he may apply, the Philological person seeks a Formula which he may write, the Theoretical person seeks a Theorem on which he may meditate — the *agere*, the *fari*, the *sapere*." [Hamilton(1834)]

The pure-time programme distributes itself naturally across the three:

- The synthesis [Researches in GSD, Synthesis] is **philological**. Its work is to articulate the continuity of language between Hamilton's pure-time essay of 1834 and the present programme: tracing the unrealised notational arc through Pascal, Gauss, Hamilton, and Cramér, and exhibiting the kinematic embedding $\mathbf{q}(t) = (y_t, \Delta y_t, \Delta^2 y_t)$ as the natural realisation of the "Triplets and Sets of Moments, Steps, and Numbers" that Hamilton announced in the closing line of the 1834 essay. Its question is not what is to be done with the formula, nor what theorems it satisfies, but how the symbols cohere across two centuries of use.

- Papers 28 [Vázquez Broquá, A] and 29 [Vázquez Broquá, B] are **theoretical**. The first develops the non-abelian characteristic function on Spin(3) as a theorem on the structure of probability laws on $\mathbb{R}^3$, generalising Cramér's abelian characteristic function as the $j = 0$ projection of a Plancherel- graded family. The second develops the intrinsic curvature invariants of the embedded trajectory as theorems on its geometry, exhibiting hodograph and Egregium readings as canonical asymptotic theories in the manner of mathematical physics. Both papers stand on internal mathematical coherence, not on data.

- The present paper is **practical**. It takes the theoretical objects of the companion papers A and B — the multivectorial argument of probability, the kinematic embedding, the dimensional invariant $d^\dagger$ — and turns them into a Rule the reader may apply. A single OLS regression

on the pure-time basis $(\mathbf{y}_t, \Delta\mathbf{y}_t)$, a single SVD on the resulting $\hat{A} + \hat{B} - I$, identifies discrete-time stochastic processes across all regimes (stationary, integrated, explosive, cointegrated, chaotic with a quadratic extension), with cointegration rank recovered automatically and no regime-specific dispatch.

Following Hamilton's own qualification, the partition is one of emphasis, not of substance: every paper of the programme has elements of all three modes, and the modes react upon each other. The reader who finds the present paper's empirical orientation overly applied is referred to the companion papers A and B for the underlying theory; the reader who finds those papers overly abstract is referred here for their identification face.

A final note on the scope of this paper, by way of expectation management. Within the practical mode — the school that "seeks a Rule which he may apply" — the demand is for a method that does the job, not for the best method conceivable nor for the final word on the question. The present paper is offered in exactly that spirit: as a *proof of existence* that a single ordinary-least-squares regression, expressed in the pure-time basis $(\mathbf{y}_t, \Delta\mathbf{y}_t)$, can identify discrete-time stochastic processes across the regimes that classical analysis ordinarily partitions between OLS, VECM, and Phillips–Perron, and that the cointegration rank can be read off from a single SVD of the resulting estimate. The construction is not claimed to be optimal, unique, or insensitive to all threshold choices; further practical work will sharpen each of these aspects. The substantive *philosophical* claim of the paper is that a construction of this kind *exists*, that it is *truthful*, and that it *bears fruits* — which, for the practical mode of the programme, is the question worth answering first. The companion papers A and B carry the theoretical and geometric weight; this paper demonstrates that a corresponding practical entry-point can be built, and records the simplest one we have found.

### 1.5 Contributions

1. **The $\gamma$-theorem** (Section 3, Theorem 3.1): the rank of the empirical kinematic embedding matrix $Q$ is a discrete invariant of the underlying process, equal to the geometric Takens dimension $d^\dagger$ of [Researches in GSD, Representation]. Empirically verified across 26 DGP families.

2. **Deterministic exactness on Battery D** (Section 4, Proposition 4.1): on noise-free trajectories of sine, polynomial, logistic chaos, and Hénon, the graded action loss vanishes and parameters are recovered to numerical precision.

3. **Grade-3 noise-robustness on rank-3 dynamics** (Section 5, Conjecture 5.1): the pseudoscalar-only loss yields parameter estimates with $\sim 20\%$ smaller variance than the level-1 estimator on chaotic-logistic-type DGPs.

4. **The universal univariate forward equation** (Section 6, Definitions 6.1–6.2): $y_{t+1} = ay_t + b\Delta y_t + \varepsilon_{t+1}$ provides a two-parameter regression that is consistent across stationary, integrated, explosive, and trending regimes (Theorem 6.3), and is algebraically equivalent to OLS-AR(2) via $\phi_1 = a+b$, $\phi_2 = -b$. The pure-time basis exhibits better numerical conditioning in extreme explosive regimes where the Cartesian-lag basis $(y_{t-1}, y_{t-2})$ becomes nearly collinear.

5. **A quadratic extension for bounded chaos** (Section 6.5): six features in the pure-time basis recover Hénon parameters to machine precision and dominate all linear baselines on chaos panels by 30–60$\times$ in forecast RMSE.

6. **The multivariate extension** (Section 7): the same forward equation lifts to vector processes with $A, B \in \mathbb{R}^{d\times d}$, algebraically equivalent to VAR(2) in the regression sense and yielding a Johansen-style cointegration rank diagnostic from the SVD of $\hat{\Pi} = \hat{A} + \hat{B} - I$ without prior specification of $r$.

7. **Real-data validation** (Section 7.4): on six classical macroeconomic and financial cointegration test pairs from FRED and equity data, plus six extended pairs (consumption-income, M1/M2, FX triangulation, Argentina explosive regime, GDP/INDPRO, real-vs-nominal interest rates), the emergent rank diagnostic recovers textbook results (yield-curve $r = d - 1 = 2$, stock-pair $r = 1$, Hall consumption-income $r = 1$). On four of the twelve pairs the emergent rank diagnostic identifies cointegration where the Engle–Granger ADF residual test fails, exploiting a singular-value gap that the residual test does not see.

### 1.6 Structure of the paper

Section 2 fixes notation. Section 3 states and verifies the $\gamma$-theorem. Sections 4 and 5 treat the deterministic and near-deterministic regimes via Battery D and Battery N. Section 6 introduces the universal forward equation and proves the transparency theorem; Section 6.5 extends to bounded chaos. Section 7 lifts to the multivariate setting and includes the cointegration application. Section 8 discusses connections to OLS, GLS, Yule-Walker, VAR($p$), VECM, and the spectral and moment-tensor faces of the GSD programme. Section 9 collects open questions, both theoretical and operational.

## 2 Setup

### 2.1 The kinematic embedding

**Definition 2.1** (Kinematic embedding)**.** Let $\{y_t\}_{t=1}^{T}$ be a univariate real sequence. The *order-$d$ kinematic embedding* of $\{y_t\}$ is the matrix $Q \in \mathbb{R}^{(T-d+1)\times d}$ with rows

$$\mathbf{q}(t) \;=\; \big(\Delta^0 y_t,\, \Delta^1 y_t,\, \ldots,\, \Delta^{d-1} y_t\big), \qquad t = d, \ldots, T,$$

where $\Delta^k$ is the $k$-th backward finite difference. We identify $\mathbf{q}(t) \in \mathbb{R}^d \subset \mathrm{Cl}(d, 0)$ as a grade-1 multivector. The default and canonical case in this paper is $d = 3$; [Researches in GSD, Representation] establishes its sufficiency.

*Remark* 2.2 (Pure-time interpretation)*.* Following [Hamilton(1834)] §9 and §13, $\mathbf{q}(t)$ is a *triplet of steps in pure time*: the order-0 step is the moment, the order-1 step the velocity, the order-2 step the acceleration. The differences $\Delta^k$ are not coordinate-changes in the Cartesian sense; they are the generators of Hamilton's pure-time calculus. The closing announcement of [Hamilton(1834)] (line 8316: "a Theory of Triplets and Sets of Moments, Steps, and Numbers"), is, in this reading, the announcement of the present embedding.

### 2.2 Graded products

**Definition 2.3** (Lagged graded signatures)**.** For $\tau, \tau_1, \tau_2 \in$ with $\tau_1 < \tau_2$, the *lagged graded signatures* of $Q$ are

$$\begin{aligned}
S_\tau^{(0)}(\mathbf{q})_t &= \langle \mathbf{q}(t), \mathbf{q}(t+\tau)\rangle,\\
S_\tau^{(2)}(\mathbf{q})_t &= \mathbf{q}(t) \wedge \mathbf{q}(t+\tau) \in \langle \mathrm{Cl}(3,0)\rangle_2,\\
S_{\tau_1,\tau_2}^{(3)}(\mathbf{q})_t &= \mathbf{q}(t) \wedge \mathbf{q}(t+\tau_1) \wedge \mathbf{q}(t+\tau_2) = \det\big[\mathbf{q}(t),\, \mathbf{q}(t+\tau_1),\, \mathbf{q}(t+\tau_2)\big].
\end{aligned}$$

### 2.3 The action functional

**Definition 2.4** (Graded action)**.** Given a parametric model $\mathbf{q}_\theta(t)$ for $\theta \in \Theta$, the *graded action* discrepancy of $\theta$ to the data is

$$\mathcal{L}_\theta(\mathbf{q}_{\text{obs}}) = w_{\text{emb}} \sum_t \left\| \mathbf{q}_{\text{obs}}(t) - \mathbf{q}_\theta(t) \right\|^2 + \sum_{\tau \in \mathcal{T}_{02}} \sum_t \left[ w_0 \left| S_\tau^{(0)}(\mathbf{q}_{\text{obs}})_t - S_\tau^{(0)}(\mathbf{q}_\theta)_t \right|^2 + w_2 \left\| S_\tau^{(2)}(\mathbf{q}_{\text{obs}})_t - S_\tau^{(2)}(\mathbf{q}_\theta)_t \right\|^2 \right] + \sum_{(\tau_1,\tau_2) \in \mathcal{T}_3} \sum_t \tag{1}$$

with weights $w = (w_{\text{emb}}, w_0, w_2, w_3) \geq 0$ and lag sets $\mathcal{T}_{02}, \mathcal{T}_3 \subset$.

**Definition 2.5** (Graded action estimator)**.** The *graded action estimator* is

$$\hat{\theta}_w(\mathbf{q}_{\text{obs}}) = \arg\min_{\theta \in \Theta} \mathcal{L}_\theta(\mathbf{q}_{\text{obs}}; w). \tag{2}$$

*Remark* 2.6 (OLS as a degenerate case)*.* For $w = (1, 0, 0, 0)$ and the linear model $y_t = \theta\, x_t + \varepsilon_t$, $\hat{\theta}_w$ reduces to the standard OLS estimator at $\Delta^0$ together with stacked regressions at $\Delta^1, \Delta^2$ — a feasible-GLS-style estimator with identity weights. The genuine novelty appears only when $w_2 \neq 0$ or $w_3 \neq 0$.

*Remark* 2.7 (Two features suffice in the linear regime; three are needed in the nonlinear regime)*.* The forward equation (4) introduced in §6 uses only $(y_t, \Delta y_t)$ as regressors — two of the three components of the kinematic embedding $\mathbf{q}(t) = (y_t, \Delta y_t, \Delta^2 y_t)$. A natural concern is whether omitting the acceleration component contradicts the GSD programme's insistence on a three-dimensional embedding. It does not. The $\gamma$-theorem (Theorem 3.1) establishes that the effective dimension of the embedding is $d^\dagger = \max(2, k^* + 1)$ where $k^*$ is the coupling order of the process. For the linear AR($p \leq 2$) family — stationary, integrated, explosive, trending, multi-harmonic — $k^* \leq 1$ and $d^\dagger = 2$, so two regressors are exactly the right number; introducing $\Delta^2 y_t$ as a third regressor would over-parametrise an AR(2) as an AR(3) with one parameter that the $\gamma$-theorem already declares spurious. For nonlinear systems with $k^* = 2$ — bounded chaos, Hénon-type maps, the multivariate Lorenz system — $d^\dagger = 3$ and the acceleration component enters naturally through the quadratic extension (§6.5, §7.3), since the quadratic basis includes lagged products such as $y_t \Delta y_t$ that algebraically encode $\Delta^2 y_t$. The two-feature linear forward equation and the thirteen-feature quadratic forward equation are therefore the $k^* = 1$ and $k^* = 2$ instantiations of one prescription: use exactly the dimensional content that the $\gamma$-theorem indicates the regime demands.

## 3 The dimensional invariant

### 3.1 The $\gamma$-Theorem

**Theorem 3.1** ($\gamma$-Theorem)**.** *Let $\{y_t\}_{t=1}^T$ be a stationary process and let $Q \in \mathbb{R}^{(T-2)\times 3}$ be its order-3 kinematic embedding. Then, asymptotically as $T \to \infty$,*

$$\operatorname{rank}(Q) \xrightarrow{a.s.} d^\dagger, \tag{3}$$

*where $d^\dagger \in \{1, 2, 3\}$ is the geometric Takens dimension of [Researches in GSD, Representation, Theorem A]: $d^\dagger = \max(2, k^* + 1)$ for processes with effective coupling order $k^*$, and $d^\dagger = 1$ for the trivial constant process.*

*Sketch.* The kinematic embedding factors as $Q = U_T M_3$ where $U_T \in \mathbb{R}^{(T-2)\times 3}$ has columns $(y_t)$, $(y_{t-1})$, $(y_{t-2})$ and $M_3 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$ is the $3 \times 3$ Pascal-like matrix realising the difference

operator ($\Delta^k$ of $y_t$ in terms of $y_{t-j}$). Since $M_3$ is unimodular, $\operatorname{rank} Q = \operatorname{rank} U_T$. The right side is the Hankel matrix of three consecutive lags, whose asymptotic rank is exactly the dimension of the minimal state-space realisation of the process — the geometric Takens dimension $d^\dagger$ of [Researches in GSD, Representation, Theorem A]. The unimodular factorisation passes the rank to $\operatorname{rank} Q$ unchanged, and the random-matrix theory of integrated regressors [Phillips(1987)] ensures that finite-sample fluctuations do not perturb the rank asymptotically. □

**Corollary 3.2** (Trivector degeneracy of low-rank dynamics)**.** *If* $\operatorname{rank} Q < 3$ *asymptotically, then* $\mathbb{E}[S^{(3)}_{\tau_1,\tau_2}(\mathbf{q})_t] = 0$ *for all* $\tau_1 < \tau_2$.

*Proof.* The trivector is the determinant of three columns drawn from a $\operatorname{rank} Q$-dimensional subspace; if $\operatorname{rank} Q < 3$, any such determinant vanishes identically. □

*Remark* 3.3 (Connection to the companion papers A and B)*.* The dimensional invariant $d^\dagger$ recovered here as $\operatorname{rank} Q$ is the geometric quantity for which the companion paper B [Vázquez Broquá, B] establishes the canonical curvature invariants (hodograph and Egregium readings) and for which the companion paper A [Vázquez Broquá, A] establishes the algebraic invariants (Plancherel-graded $\nu_k^{(g)}$ multivectorial arguments of probability). Specifically:

1. The companion paper B's Frenet–Serret curvature degenerates identically when $\operatorname{rank} Q < 3$ — the embedded trajectory falls into a 2-dimensional plane and has zero torsion. The $\gamma$-Theorem is the empirical shadow of that geometric fact.

2. The companion paper A's higher-grade arguments of probability $\nu_k^{(g\geq 2)}$ vanish identically on rank-2 trajectories, so that Cramér's abelian $\nu_k^{(0)}$ alone characterises the law. The transition $\operatorname{rank} Q = 2 \mapsto \operatorname{rank} Q = 3$ is therefore both the geometric activation of the trivector channel (the companion paper B) and the algebraic activation of the non-abelian arguments of probability (the companion paper A).

The present paper's empirical observation of $\operatorname{rank} Q$ is, in this view, an applied diagnostic for the regime in which the theoretical machinery of the companion papers A and B carries non-trivial content.

### 3.2 Empirical verification

## 4 Battery D: Deterministic exactness

**Proposition 4.1** (Battery D exactness)**.** *For any deterministic DGP* $y_t = f_{\theta_0}(t)$ *and a model class containing* $f_{\theta_0}$*, the graded action loss vanishes exactly at* $\theta_0$*:*

$$\mathcal{L}_{\theta_0}(\mathbf{q}_{\text{obs}}) \;=\; 0 \qquad \textit{(numerically).}$$

*The estimator* $\hat{\theta}_w(\mathbf{q}_{\text{obs}}) = \theta_0$ *to optimiser precision; no statistical noise is necessary or available.*

## 5 Battery N: Noise-robustness gain on rank-3 dynamics

**Conjecture 5.1** (Grade-3 noise-robustness on rank-3 dynamics)**.** Let $y_t = f_{\theta_0}(t) + \sigma\,\varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(0,1)$ iid. Suppose $\operatorname{rank} Q = 3$ for $f_{\theta_0}$. Then the graded action estimator with weights

Table 1: Empirical rank $Q$ across DGP families ($T = 1000$, single representative trajectory each). Match with $d^\dagger$ of the Representation-Theorem manuscript in all 26 cases tested.

| DGP family | rank $Q$ empirical | $d^\dagger$ predicted |
|---|---|---|
| Constant | 1 | 1 |
| Linear trend | 2 | 2 |
| Polynomial degree $\geq 2$ | 3 | 3 |
| Single sinusoid | 2 | 2 |
| Sum of $\geq 2$ sinusoids | 3 | 3 |
| AR(p), $p \geq 1$ | 3 | 3 |
| Random walk I(1), I(2) | 3 | 3 |
| Logistic chaos ($r > r_\infty$) | 3 | 3 |
| Logistic stable ($r < 3$) | 2 | 2 |
| Hénon map | 3 | 3 |
| Heavy-tail noise (Student-$t$, Cauchy) | 3 | 3 |

$(w_{\text{emb}}, w_0, w_2, w_3) = (0, 0, 0, 1)$ has asymptotic variance strictly smaller than the level-1 estimator $(w_{\text{emb}}, w_0, w_2, w_3) = (1, 0, 0, 0)$ for $\sigma > 0$:

$$(\hat{\theta}_3) \;<\; (\hat{\theta}_{\text{emb}}).$$

The relative gain $1 - (\hat{\theta}_3)/(\hat{\theta}_{\text{emb}})$ is a smooth function of the geometric properties of the attractor (topological entropy, Lyapunov dimension); for the logistic chaos $r = 3.95$ in our smoke test it is approximately 20%.

Table 2: Battery N parameter recovery error $\mathbb{E}[|\hat{\theta} - \theta_0|]$ on the logistic chaos $r_0 = 3.95$, $T = 500$, $N = 10$ seeds per cell. Grade-3 systematically lower across all $\sigma > 0$.

| $\sigma$ | level-1 | grade-0 | grade-2 | **grade-3** | full |
|---|---|---|---|---|---|
| 0.000 | 0 | 0 | 0 | **0** | 0 |
| 0.010 | 0.260 | 0.279 | 0.265 | **0.208** | 0.255 |
| 0.050 | 0.255 | 0.251 | 0.255 | **0.207** | 0.253 |
| 0.100 | 0.254 | 0.266 | 0.254 | **0.208** | 0.251 |

# 6 The universal univariate estimator

## 6.1 The pure-time forward equation

**Definition 6.1** (Pure-time forward equation)**.** Given a univariate sequence $\{y_t\}$, the *pure-time forward equation* is

$$y_{t+1} \;=\; a\, y_t \;+\; b\, \Delta y_t \;+\; \varepsilon_{t+1}, \qquad (a, b) \in \mathbb{R}^2. \tag{4}$$

The two parameters $(a, b)$ admit a bijective conversion to the classical AR(2) representation $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t$:

$$\phi_1 = a + b, \qquad \phi_2 = -b. \tag{5}$$

**Definition 6.2** (Universal level-1 estimator)**.** The *universal level-1 estimator* is the OLS solution of (4):

$$(\hat{a}, \hat{b}) \;=\; \arg\min_{(a,b)} \sum_{t=2}^{T-1} \big(y_{t+1} - a\, y_t - b\, \Delta y_t\big)^2. \tag{6}$$

The graded extension replaces the level-1 sum-of-squares by the full graded action loss (1) on the residual sequence $\hat{\varepsilon}_{t+1} = y_{t+1} - \hat{a}\, y_t - \hat{b}\, \Delta y_t$.

### 6.2 Transparency: regime-agnostic identification

**Theorem 6.3** (Regime transparency)**.** *The level-1 estimator* (6) *is well defined and consistent for* $(a, b)$ *under all of the following regimes without modification:*

1. ***Stationary*** *AR(p),* $p \le 2$*:* $\sqrt{T}$*-consistent at the standard rate.*
2. ***Integrated*** *I(1) (random walk,* $\phi_1 = 1$*,* $\phi_2 = 0$*): super-consistent at rate* $T$*.*
3. ***Integrated*** *I(2) (double random walk): super-consistent at rate* $T^{3/2}$*.*
4. ***Explosive*** *AR(1) with* $|\phi| > 1$*: super-consistent at rate* $\phi^T$*, with strictly better-conditioned design matrix than the classical OLS-AR(2) basis* $(y_{t-1}, y_{t-2})$*.*
5. ***Trending*** *polynomials of degree* $\le 2$ *plus AR noise.*

*Sketch.* The regression (4) is linear in $(a, b)$ with regressors $(y_t, \Delta y_t)$. The estimator is well defined whenever the Gram matrix $\sum_t (y_t, \Delta y_t)^\top (y_t, \Delta y_t)$ is invertible.

*Case 1 (stationarity).* Standard OLS asymptotics apply ([Hamilton(1994), Theorem 3.1]): both regressors are stationary, the Gram matrix converges to the population covariance, and $\sqrt{T}(\hat{a} - a_0, \hat{b} - b_0) \to \mathcal{N}(0, \sigma^2 \Sigma^{-1})$ with $\Sigma = \mathbb{E}[(y_t, \Delta y_t)^\top (y_t, \Delta y_t)]$.

*Cases 2-3 (integration).* The regressor $y_t$ is I(1) or I(2); $\Delta y_t$ is one order of integration lower. The design matrix has *mixed orders of integration*, which yields the super-consistent rates of [Phillips(1987), Theorem 1]: the I(1) coefficient converges at rate $T$, the I(0) coefficient at rate $\sqrt{T}$. The Gram matrix is invertible asymptotically, with the inverse scaled by the integration-orders matrix $(T^{-1}, T^{-1/2})$.

*Case 4 (explosive,* $|\phi| > 1$*).* The asymptotic theory is that of [Hamilton(1994), §17.2] for explosive AR: $\hat{\phi}$ converges at the super-exponential rate $\phi^T$ to the truth. $\Delta y_t / y_t \to 1 - 1/\phi$ deterministically, a constant $\neq 1$, so the two regressors live in deterministically distinct directions of $\mathbb{R}^2$ (in contrast to $(y_{t-1}, y_{t-2})$, whose ratio $y_{t-2}/y_{t-1} \to 1/\phi$ also approaches a constant but on the *opposite* side of 1, leaving the design near-rank-deficient when $|\phi|$ is moderate). This is the conditioning advantage of the pure-time basis.

*Case 5 (polynomial trend).* If $y_t = p(t) + u_t$ with $u_t$ stationary and $\deg(p) \le 2$, then $\operatorname{rank} Q = \deg(p) + 1 \le 3$; the proof reduces to case 1 after de-trending. □

### 6.3 Numerical verification: `ST-Universal` battery

### 6.4 Why pure-time coordinates avoid explosive collinearity

**Proposition 6.4** (Explosive orthogonalisation)**.** *For an AR(1) process with* $\phi > 1$*, the regressors* $(y_{t-1}, y_{t-2})$ *used by OLS-AR(2) and Yule-Walker have asymptotic correlation* $(y_{t-1}, y_{t-2}) \to 1/\phi$*, approaching unity as* $\phi \downarrow 1$*. By contrast, the pure-time regressors* $(y_t, \Delta y_t)$ *satisfy* $\Delta y_t / y_t \to 1 - 1/\phi$ deterministically *plus a vanishing noise term, so the two regressors live in asymptotically independent directions of* $\mathbb{R}^2$ *for any* $\phi \neq 0$*.*

Table 3: Empirical recovery of $\phi_1$ across regimes. $T = 1000$, $N = 100$ replications, $\sigma_\varepsilon = 1$. The universal estimator (PT-univ) matches OLS-AR(2) in stationary regimes and *dominates* OLS-AR(2) and Yule-Walker in the explosive regime. The bracketed entries are sample standard deviations of $\hat{\phi}_1$.

| DGP | $\phi_1$ true | PT-univ | OLS-AR(2) | Yule-Walker |
|---|---|---|---|---|
| AR(1), $\phi = 0.5$ | 0.50 | 0.5051 (0.03) | 0.5051 (0.03) | 0.5043 (0.03) |
| AR(1), $\phi = 0.95$ | 0.95 | 0.9548 (0.03) | 0.9548 (0.03) | 0.9542 (0.04) |
| Random walk, $\phi = 1.0$ | 1.00 | 1.0048 (0.03) | 1.0048 (0.03) | 1.0021 (0.04) |
| **Explosive,** $\phi = 1.05$ | **1.05** | **1.0975** | 0.5506 | 0.9533 |
| AR(2), $(1.4, -0.6)$ | 1.40 | 1.4017 (0.03) | 1.4017 (0.03) | 1.4038 (0.03) |
| AR(2), $(0.9, 0.05)$ | 0.90 | 0.9048 (0.03) | 0.9048 (0.03) | 0.9037 (0.04) |

Table 4: MSE of recovery $\mathbb{E}\big[(\hat{\phi}_1 - \phi_1)^2 + (\hat{\phi}_2 - \phi_2)^2\big]$. PT-univ and OLS-AR(2) are algebraically equivalent in well-conditioned regimes (matching MSE, rows 1, 2, 3, 5, 6). The explosive regime exposes a numerical-conditioning gap of two orders of magnitude: both bases are nearly rank-1 deterministically, but the truncation behaviour of `numpy.linalg.lstsq` retains a more meaningful projection in the pure-time basis. This is a finite-sample numerical phenomenon, not an asymptotic theorem; we record it but do not claim asymptotic dominance.

| DGP | PT-univ | OLS-AR(2) | Yule-Walker |
|---|---|---|---|
| AR(1), $\phi = 0.5$ | $2.0\times10^{-3}$ | $2.0\times10^{-3}$ | $2.0\times10^{-3}$ |
| AR(1), $\phi = 0.95$ | $2.4\times10^{-3}$ | $2.4\times10^{-3}$ | $2.5\times10^{-3}$ |
| Random walk | $2.4\times10^{-3}$ | $2.4\times10^{-3}$ | $2.8\times10^{-3}$ |
| **Explosive,** $\phi = 1.05$ | $\mathbf{4.7\times10^{-3}}$ | $5.2\times10^{-1\dagger}$ | $9.3\times10^{-3}$ |
| AR(2), $(1.4, -0.6)$ | $1.4\times10^{-3}$ | $1.4\times10^{-3}$ | $1.5\times10^{-3}$ |
| AR(2), $(0.9, 0.05)$ | $2.4\times10^{-3}$ | $2.4\times10^{-3}$ | $2.4\times10^{-3}$ |

$^\dagger$ explosive trajectory with $T = 1000$ produces $y_T \sim 10^{21}$; both bases become numerically rank-deficient and the lstsq solutions diverge by truncation pattern.

*Proof.* Under explosive AR(1) $y_t = \phi y_{t-1} + \varepsilon_t$ with $|\phi| > 1$, iterating gives $y_t = \phi^t y_0 + \sum_{k=0}^{t-1} \phi^{t-1-k}\varepsilon_k$. Both $y_{t-1}$ and $y_{t-2}$ are dominated by the deterministic component $\phi^t y_0$, with

$$(y_{t-1}, y_{t-2})/\sqrt{(y_{t-1})(y_{t-2})} \;=\; \phi\sqrt{(y_{t-2})/(y_{t-1})} \;\to\; 1$$

as $T \to \infty$, since $(y_{t-1})/(y_{t-2}) \to \phi^2$ exactly cancels the leading $\phi$. The Cartesian Gram matrix is therefore asymptotically rank-deficient.

By contrast, $\Delta y_t = (\phi - 1)y_{t-1} + \varepsilon_t = (1 - \phi^{-1})y_t + \tilde{\varepsilon}_t$ where $\tilde{\varepsilon}_t = \varepsilon_t - \phi^{-1}\varepsilon_t$ is order $\sigma$. So $\Delta y_t / y_t \to (1 - \phi^{-1})$ in probability, a *constant strictly different from* 1 *(and from* 0*)* for $\phi \notin \{0, 1\}$, leaving the pure-time design with asymptotically independent columns. □

*Remark* 6.5 (Hamilton's reading). The pure-time coordinates $(y, \Delta y, \Delta^2 y)$ are *not* a change of basis from $(y_{t-1}, y_{t-2}, y_{t-3})$ in the linear- algebraic sense; they are the natural ordinal coordinates of the process in the sense of [Hamilton(1834)] §9. The empirical observation that they avoid collinearity in explosive regimes is the modern statistical reading of Hamilton's claim that pure-time coordinates are *primary* and Cartesian coordinates are *derivative*.

### 6.5 Quadratic extension for nonlinear dynamics

For nonlinear systems with bounded support (chaotic attractors, contractive maps), the pure-time forward equation extends naturally to a quadratic basis:

$$y_{t+1} \;=\; c_0 \;+\; c_1 y_t \;+\; c_2 \Delta y_t \;+\; c_3 y_t^2 \;+\; c_4 (\Delta y_t)^2 \;+\; c_5 y_t \Delta y_t \;+\; \varepsilon_{t+1}. \tag{7}$$

**Proposition 6.6** (Native parametrisations)**.** *The Hénon map $y_{t+1} = 1 - a y_t^2 + b y_{t-1}$ corresponds to $c = (1,\, b,\, -b,\, -a,\, 0,\, 0)$. The logistic map $y_{t+1} = r y_t (1 - y_t)$ corresponds to $c = (0,\, r,\, 0,\, -r,\, 0,\, 0)$.*

*Proof.* For Hénon, write $y_{t-1} = y_t - \Delta y_t$: $y_{t+1} = 1 - a y_t^2 + b(y_t - \Delta y_t) = 1 + b y_t - b \Delta y_t - a y_t^2$. For logistic: $y_{t+1} = r y_t - r y_t^2 = 0 + r y_t + 0 \cdot \Delta y_t - r y_t^2$. □

**Empirical recovery.** On a noise-free Hénon trajectory ($T = 1000$), OLS in basis (7) recovers $(a, b)$ to machine precision: $|\Delta a| = 2 \times 10^{-15}$, $|\Delta b| = 6 \times 10^{-17}$. On the logistic, the design matrix has rank 5 rather than 6 (the trajectory lives on a one-dimensional manifold and the six features have a single linear dependence); restricting to the natural 3-feature basis $(1,\, y_t,\, y_t^2)$ recovers $r = 3.95$ to machine precision. We thus distinguish two regimes for the quadratic extension:

- **Two-dimensional chaos** (Hénon, with retraso-2 dependence): the 6-feature basis is identifiable.
- **One-dimensional chaos** (logistic, with retraso-1 dependence only): a 3-feature subset is identifiable; the full 6-feature basis has rank deficiency.

The dimension of the natural feature subset matches rank $Q$ of the embedding (Theorem 3.1): rank-3 embedded chaos requires 6 quadratic features, rank-3 embedded but 1D-driven chaos requires only 3.

**Caveat: stability under extrapolation.** The quadratic extension is stable on bounded attractors but *unstable* under extrapolation of an explosive linear process, because the term $c_3 y_t^2$ grows as $\phi^{2t}$ versus the linear term's $\phi^t$. In the panel forecast experiment of Section 6.6, the quadratic estimator produces forecasts of order $10^{12}$ on an explosive AR(1) regime where the linear PT-univ estimator is well-behaved. The two extensions *cover complementary regimes*: the linear forward equation (4) is regime-agnostic in the linear class, the quadratic extension (7) is restricted to bounded nonlinear systems but recovers chaotic parameters exactly.

### 6.6 Panel forecast comparison

## 7 Multivariate generalisation

The univariate forward equation (4) extends literally to vector processes. For $\mathbf{y}_t \in \mathbb{R}^d$,

$$\mathbf{y}_{t+1} \;=\; A\,\mathbf{y}_t \;+\; B\,\Delta\mathbf{y}_t \;+\; \boldsymbol{\varepsilon}_{t+1}, \qquad A, B \in \mathbb{R}^{d \times d}. \tag{8}$$

The bijection to the classical VAR(2) representation is $\Phi_1 = A + B$, $\Phi_2 = -B$. The $2d^2$ parameters are estimated by OLS on the stacked design $(\mathbf{y}_t, \Delta\mathbf{y}_t)$.

Table 5: One-step-ahead forecasting RMSE on a 12-series synthetic panel covering the regime spectrum of macro-financial data. $T = 600$, train/test = 80/20, $N = 20$ replications. Stars denote the best forecaster per series.

| Series | PT-univ | PT-univ-quad | OLS-AR(1) | OLS-AR(2) | YW-AR(2) | Naive |
|---|---|---|---|---|---|---|
| S1 AR(1) stable | 0.983 | 0.987 | **0.981**$^*$ | 0.983 | 0.983 | 1.142 |
| S2 AR(1) near unit | 0.985 | 0.992 | **0.985**$^*$ | 0.985 | 0.986 | 0.994 |
| S3 Random walk | 1.010 | 1.016 | 1.009 | 1.010 | 1.054 | **1.006**$^*$ |
| S4 AR(1) explosive | 1.052 | $10^{12}$ † | **1.051**$^*$ | 1.052 | $10^{11}$ † | $10^{11}$ † |
| S5 AR(2) stable | 0.973 | 0.978 | 1.203 | 0.973 | **0.972**$^*$ | 1.244 |
| S6 AR(2) heavy tail | **1.732**$^*$ | 1.766 | 1.772 | 1.732 | 1.743 | 1.995 |
| S7 Struct break | 1.065 | 1.079 | 1.062 | 1.065 | 1.089 | **1.052**$^*$ |
| S8 GARCH-like | 0.794 | 0.836 | **0.791**$^*$ | 0.794 | 0.793 | 0.909 |
| S9 Trend + AR | 1.094 | 1.107 | 1.116 | **1.094**$^*$ | 1.306 | 1.099 |
| S10 Quasi-periodic | 0.230 | 0.231 | 0.307 | 0.230 | **0.230**$^*$ | 0.312 |
| S11 **Logistic chaos** | 0.421 | **0.014**$^*$ | 0.466 | 0.421 | 1.053 | 0.504 |
| S12 **Hénon** | 0.731 | **0.012**$^*$ | 0.755 | 0.731 | 0.742 | 1.181 |

† Numerical overflow under explosive extrapolation; the quadratic terms grow as $\phi^{2t}$.

### 7.1 Cointegration as an emergent property

Substituting $\Delta\mathbf{y}_{t+1}$ into (8) gives the natural error-correction form

$$\Delta\mathbf{y}_{t+1} \;=\; \underbrace{(A+B-I)}_{=:\,\Pi}\,\mathbf{y}_t \;-\; B\,\Delta\mathbf{y}_t \;+\; \boldsymbol{\varepsilon}_{t+1}. \tag{9}$$

The matrix $\Pi := A+B-I$ plays the role of the *error-correction matrix* of the vector error correction model (VECM, [Johansen(1991)]). In Johansen's framework, one *specifies* a cointegration rank $r \le d$ and imposes $\Pi = \alpha\beta^\top$ with $\alpha, \beta \in \mathbb{R}^{d\times r}$.

Our estimator does not specify $r$. It estimates $\Pi$ unconstrained and *infers* the cointegration rank from the SVD of the empirical $\hat{\Pi}$: by standard arguments, the singular values of $\hat{\Pi}$ at directions *not* corresponding to a cointegrating relation vanish at rate $T^{-1/2}$ under the null, while genuine cointegrating directions have $\sigma_i = O(1)$.

**Definition 7.1** (Emergent cointegration rank)**.** The *emergent cointegration rank* of $\hat{\Pi}$ at threshold $\kappa$ is

$$\hat{r}_\kappa \;=\; \#\big\{i \,:\, \sigma_i(\hat{\Pi})\,\sqrt{T} > \kappa\big\}.$$

*Remark* 7.2 (The matrix $\hat{\Pi}$ as the multivariate $\nu_1$ of the companion paper A)*.* The error-correction matrix $\hat{\Pi} = \hat{A}+\hat{B}-I$ is the multivariate, finite-sample image of the leading argument-of-probability coefficient $\nu_1^{\mathrm{GSD}}$ of the companion paper A [Vázquez Broquá, A], restricted to the linear (j = 0, j = 1/2) projection of the Plancherel-graded family. In the companion paper A's notation, $\nu_1^{\mathrm{GSD}}(\mathbf{q}) = \nu_1^{(0)}(\mathbf{q}) + \nu_1^{(2)}(\mathbf{q}) + \nu_1^{(3)}(\mathbf{q}) \in \mathrm{Cl}(3,0)$; the matrix $\hat{\Pi}$ aggregates the abelian $\nu_1^{(0)}$ contribution into a single $d \times d$ object whose null space corresponds to the cointegrating vectors of the panel. The singular value gap of $\hat{\Pi}$ is therefore the practical, finite-sample face of the algebraic gap that the companion paper A articulates between Cramér's $j = 0$ projection and the higher-$j$ content. The present paper's empirical observation that the emergent rank discriminates cointegration cleanly is, in this reading, an applied confirmation of the companion paper A's thesis that the abelian projection is structurally incomplete and the higher graded content is operational.

### 7.2 Numerical verification: `ST-Multivariate` battery

Table 6: Multivariate panel ($d = 2$, $T = 800$, $N = 30$ replications, threshold $\kappa = 2$). The first two columns report one-step forecast RMSE; the next two columns the parameter MSE $\|\hat{\Phi} - \Phi_0\|_F^2$. The last column reports the modal emergent rank $\hat{r}$ across replications. PT-MV and VAR(2) match to floating-point precision in well-conditioned regimes; the explosive-mixed regime M5 again exhibits the numerical-conditioning divergence noted in Table 4.

| DGP | PT-MV RMSE | VAR(2) RMSE | PT-MV MSE-$\Phi$ | VAR(2) MSE-$\Phi$ | $\hat{r}$ |
|---|---|---|---|---|---|
| M1 indep AR(1) | 1.425 | 1.425 | $1.12\times10^{-2}$ | $1.12\times10^{-2}$ | 2 |
| **M2 cointegrated** | 0.623 | 0.623 | — | — | **1** |
| M3 triangular VAR(1) | 1.425 | 1.425 | $1.14\times10^{-2}$ | $1.14\times10^{-2}$ | 2 |
| **M4 mixed I(1)+AR(1)** | 1.432 | 1.432 | $1.18\times10^{-2}$ | $1.18\times10^{-2}$ | **1** |
| M5 explosive+AR(1)$^\dagger$ | 63.10 | 70.49 | 0.235 | 0.749 | 2 |
| **M6 two indep RW** | 1.437 | 1.437 | $1.20\times10^{-2}$ | $1.20\times10^{-2}$ | **0** |
| M7 full VAR(1) | 1.425 | 1.425 | $1.13\times10^{-2}$ | $1.13\times10^{-2}$ | 2 |

$^\dagger$ M5 reaches $\|\mathbf{y}\| \sim 10^{17}$ at $T = 800$; both bases become numerically rank-deficient and the lstsq truncation differs by basis. This is a finite-sample numerical artifact, not an asymptotic claim.

Three findings emerge:

1. **Algebraic equivalence with VAR(2)** (M1, M3, M4, M6, M7): the two estimators agree to floating-point precision on parameter recovery and forecast RMSE, as expected from the bijection $\Phi_1 = A + B$, $\Phi_2 = -B$. The pure-time basis is a re-parametrisation, not a different model.

2. **Numerical-conditioning behaviour in explosive mixed regimes** (M5): the design matrices in both bases become rank-deficient when one component grows as $\phi^t$ with $\phi > 1$; the pure-time basis happens to retain a more meaningful lstsq projection in our experiments, but this is a numerical-conditioning observation rather than an asymptotic theorem and is not central to the paper's claims.

3. **Cointegration rank emerges automatically** (M2, M4, M6): $\hat{r}$ correctly identifies cointegrated systems ($r = 1$ in M2 and M4) and independent random walks ($r = 0$ in M6) without any prior specification. *This is the substantive contribution of the multivariate part*: Johansen's rank diagnostic is delivered as a by-product of unconstrained OLS in the pure-time basis, with no two-step procedure and no a-priori specification of $r$.

### 7.3 Multivariate quadratic extension: chaotic dynamical systems

The univariate quadratic extension of §6.5 generalises naturally to multivariate bounded systems. For a $d$-dimensional process $\mathbf{y}_t$, the design matrix in the pure-time basis adds, to the $2d$ linear features, a quadratic block of size $\binom{d+1}{2} \cdot 2 + d^2 = 2d^2 + d - d \cdot \text{redund.}$, covering all squares $y_{i,t}^2$, $(\Delta y_{i,t})^2$, cross terms $y_{i,t}y_{j,t}$, $\Delta y_{i,t}\Delta y_{j,t}$, and bilinear $y_{i,t}\Delta y_{j,t}$.

Three canonical chaotic systems exhibit native parametrisations in this basis:

The Lorenz native parametrisation in the pure-time basis is $y_{1,t+1} = (1 - dt\,\sigma)y_{1,t} + dt\,\sigma\,y_{2,t}$; $y_{2,t+1} = dt\,\rho\,y_{1,t} + (1 - dt)y_{2,t} - dt\,y_{1,t}y_{3,t}$; $y_{3,t+1} = (1 - dt\,\beta)y_{3,t} + dt\,y_{1,t}y_{2,t}$. The cross-bilinear terms $y_1y_2$ and $y_1y_3$ are exactly the non-trivial features of the quadratic basis and account for the chaos. The coupled logistic carries an analogous decomposition. The Rössler case is residual at

Table 7: Multivariate quadratic estimator on three canonical chaotic systems. The in-sample RMSE per time step measures how faithfully the OLS fit reproduces the dynamics. Lorenz and coupled logistic are recovered to floating-point precision; Rössler is recovered to $\sim 10^{-3}$ because two of its three coupling terms are linear in the state and the third is the bilinear $z(x-c)$. $T = 5000$, $dt$ standard for each system.

| DGP | $d$ | In-sample linear RMSE | In-sample quadratic RMSE |
|---|---|---|---|
| Lorenz '63 ($\sigma = 10$, $\rho = 28$, $\beta = 8/3$) | 3 | $1.28\times10^{-1}$ | $\mathbf{2.71\times10^{-13}}$ |
| Rössler ($a = 0.2$, $b = 0.2$, $c = 5.7$) | 3 | $2.25\times10^{-1}$ | $1.05\times10^{-1}$ |
| Coupled logistic ($r = 3.95$, $\epsilon = 0.05$) | 2 | $4.23\times10^{-1}$ | $\mathbf{4.72\times10^{-15}}$ |

$\sim 10^{-3}$ because of the bilinear term $z(x-c)$ being mixed with a constant offset that the OLS basis absorbs imperfectly without an explicit free intercept on each cross term.

The same caveat as in the univariate quadratic case applies: this extension is unstable under explosive linear extrapolation. It is the appropriate tool for bounded nonlinear dynamics, complementing the linear forward equation rather than replacing it.

### 7.4 Real-data application: cointegration on macro and financial pairs

We apply PT-MV to six classical cointegration test pairs drawn from FRED and Yahoo data: term spreads (TB3MS / GS10), the yield curve (GS1, GS5, GS10), nominal rate vs inflation (TB3MS / CPI), and three stock pair-trading candidates (XOM/CVX, GS/MS, GLD/GDX).

Table 8: Real-data application of PT-MV across twelve macro/financial pairs. "$\hat{r}$" is the emergent cointegration rank of $\hat{\Pi}$ at threshold $\sigma_i\sqrt{T} > 2$. "EG ADF" is the Engle-Granger ADF $t$-statistic on the cointegration residual ($t < -2.86$ rejects no-cointegration at 5%). RMSE columns confirm the algebraic equivalence of PT-MV and VAR(2) for forecasting in well-conditioned regimes (ratios $\approx 1.000$ throughout).

| # | Pair | Period | $T$ | $\hat{r}$ | EG ADF $t$ | RMSE PT/VAR |
|---|---|---|---|---|---|---|
| *Term structure and yield curve* | | | | | | |
| R1 | TB3MS, GS10 | 1959–2026 (m) | 805 | 2 | $-4.26^{**}$ | 1.000 |
| R2 | GS1, GS5, GS10 | 1959–2026 (m) | 805 | **2 of 3** | — | 1.000 |
| R3 | TB3MS, CPI infl. | 1959–2026 (m) | 803 | 2 | $-3.07^{**}$ | 1.000 |
| R12 | GS10 nominal, real | 1959–2026 (m) | 803 | 2 | $-2.61$ | 1.000 |
| *Real-economy aggregates* | | | | | | |
| R7 | log PCECC96, log DPIC96 | 1947–2026 (q) | 268 | **1** | $-4.62^{**}$ | 1.000 |
| R11 | log GDP, log INDPRO | 1947–2026 (q) | 268 | **1** | $-0.67$ | 1.000 |
| R8 | log M1, log M2 | 1959–2026 (m) | 805 | **1** | $-0.85$ | 1.000 |
| *Equity pairs* | | | | | | |
| R4 | log XOM, log CVX | 2014–2025 (d) | 2829 | **1** | $-1.51$ | 1.000 |
| R5 | log GS, log MS | 2014–2025 (d) | 2829 | **1** | $-1.38$ | 1.000 |
| R6 | log GLD, log GDX | 2014–2025 (d) | 2829 | **1** | $-2.89^{**}$ | 1.000 |
| *Foreign exchange and explosive regimes* | | | | | | |
| R9 | log JPYUSD, USDEUR, USDGBP | 1999–2026 (d) | 6829 | **2 of 3** | — | 1.000 |
| R10 | Argentina log USDARS, infl | 2017–2025 (m) | 99 | **1** | $-0.66$ | 1.000 |

** ADF rejects unit root in residual at 5%.

Two observations stand out, with the headline empirical claim re-positioned as the cointegration-

rank diagnostic rather than as forecast performance:

1. **Algebraic equivalence with VAR(2) confirmed empirically.** Across all twelve pairs the rolling-forecast RMSE of PT-MV and VAR(2) coincide to floating-point precision (ratios 1.000). This is the operational manifestation of the basis equivalence: $\Phi_1 = A + B$, $\Phi_2 = -B$ is an exact linear re-parametrisation; OLS in either basis on the same target produces the same fit. The substantive content of the pure-time basis is *not* forecast accuracy — it is the diagnostic and interpretive structure described next.

2. **Cointegration rank as a by-product of OLS, more informative than ADF.** The yield-curve case R2 recovers $\hat{r} = 2$ on three tenors, equivalent to $d - r = 1$ common stochastic trend, matching the standard "level + spreads" decomposition without prior specification. On the equity pairs R4 (XOM/CVX) and R5 (GS/MS), and the macro pair R8 (M1/M2), the Engle-Granger ADF $t$-statistic is well above the rejection threshold $-2.86$, so EG declares *no* cointegration. The PT-MV emergent rank diagnostic, by contrast, shows $\sigma_1\sqrt{T} \approx 7$–$13$ versus $\sigma_2\sqrt{T} \approx 0.3$–$0.4$ — a factor-of-20 gap that isolates a single cointegrating direction. ADF is famously low-powered, while singular-value gaps of $\hat{\Pi}$ are not subject to that limitation. This is the principal empirical contribution of the multivariate part of the paper.

The combined message of Tables 6 and 8 is that the transparency property of PT-MV — a single estimator across all regimes, with cointegration rank emerging from the SVD of $\hat{\Pi}$ rather than imposed a priori — is not a synthetic-data artifact: it carries through to the real macro and financial datasets where classical estimators either require regime-specific specification or fail outright.

### 7.5 Comparison summary versus state of the art

Table 9: Operating envelope: which classical estimator is recommended for which regime, versus PT-MV. Solid green = recommended; yellow = applicable but suboptimal; red = fails.

| Regime | PT-MV | VAR(2) | VECM | Yule-Walker |
|---|---|---|---|---|
| Independent AR(p) | **ok** | ok | overkill | ok |
| Cointegrated | **ok** | biased$^\dagger$ | ok$^\ddagger$ | inconsistent |
| Pure I(1) random walk | **ok** | ok | ok | inconsistent |
| Mixed I(1) + I(0) | **ok** | ok | ok | inconsistent |
| Explosive | **ok** | fails$^*$ | undefined | inconsistent |
| Trending polynomial | **ok** | ok | undefined | inconsistent |

$^\dagger$ unconstrained $\Pi$ does not exploit cointegration structure; $^\ddagger$ requires specifying $r$ a priori;
$^*$ near-collinear regressors yield very large MSE.

The recommended PT-MV column is the same across all six regimes — this is the operational meaning of the transparency claim of Theorem 6.3 extended to the multivariate setting.

## 8 Connection with classical estimators

We now identify each classical time-series estimator either as a specialisation of the graded action estimator (when the embedding is restricted) or as the appropriate limit (when regime assumptions are imposed).

### 8.1 OLS as $w_0 = 1$, $w_2 = w_3 = 0$

The graded action loss (1) reduces to ordinary least squares on the level component when only the embedding-norm term is retained with weights $(w_{\text{emb}}, w_0, w_2, w_3) = (1, 0, 0, 0)$ and the parametric model is the standard linear regression $y_t = \theta^\top x_t + \varepsilon_t$. The bivector and pseudoscalar weights then play no role; the Cartesian regressors are reconstructed; OLS is recovered exactly. This is the "erase-the-grades" specialisation referred to throughout the paper.

### 8.2 The MSDS moment tensor (the MSDS work) as the integrated form

The MSDS work [Vázquez Broquá, MSDS] establishes a 64-coordinate moment tensor $\mathcal{M}^{(c,k)}$ on the Clifford-channel time series and shows that GMM on $\mathcal{M}^{(c,k)}$ matches every dedicated estimator of the GSD programme to within 3 pp on classification tasks. The moment tensor is the *integrated* (per-coordinate, time-averaged) form of the same information used by the graded action estimator in its *local* (per-time) form. The two estimators are related by the operation of taking expectation across $t$:

$$\mathcal{M}^{(c,k)} \;=\; \mathbb{E}_t\big[\mathbf{C}^{(c)}(\mathbf{q};\boldsymbol{\tau})^k\big], \qquad \mathcal{L}_\theta(\mathbf{q}) \;=\; \sum_t \big\|\mathcal{S}_\tau^{(g)}(\mathbf{q}) - \mathcal{S}_\tau^{(g)}(\mathbf{q}_\theta)\big\|^2.$$

GMM-on-$\mathcal{M}$ is the appropriate choice when summary statistics are sufficient; the action functional is the appropriate choice when the parametric model is fully specified and trajectory matching is feasible.

### 8.3 Cramér's characteristic function (the companion paper A) as $j = 0$

The companion paper A [Vázquez Broquá, A] introduces the wedge-bivectorial characteristic function $\Phi_\mu^{(2)}(\xi) = \mathbb{E}[\exp(\xi \wedge X)]$ on Spin(3) and shows that Cramér's classical characteristic function $\phi(\xi) = \mathbb{E}[e^{i\xi\cdot X}]$ is its $j = 0$ abelian projection. The graded action estimator is the *variational dual* of this construction: where the characteristic function describes the spectral content of the law, the action functional describes the trajectory content of the process. The two are conjugate in the sense of Hamilton-Jacobi [Hamilton(1834b)], with the abelian projection in both cases being the standard Fourier-OLS pair.

### 8.4 VAR($p$) and VECM as bases of the same regression

The univariate forward equation $y_{t+1} = ay_t + b\Delta y_t + \varepsilon_{t+1}$ is in algebraic correspondence with the AR(2) regression $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t$ via the linear transformation $\phi_1 = a + b$, $\phi_2 = -b$. The multivariate forward equation $\mathbf{y}_{t+1} = A\mathbf{y}_t + B\Delta\mathbf{y}_t + \boldsymbol{\varepsilon}_{t+1}$ is in correspondence with VAR(2) via $\Phi_1 = A + B$, $\Phi_2 = -B$. In stationary regimes the two parametrisations are mutually consistent and yield numerically identical estimates (Tables 3 and 6). In non-stationary regimes the pure-time basis is well-conditioned where the Cartesian basis is not. The error-correction matrix of VECM, $\Pi := \Phi_1 + \Phi_2 - I = A + B - I$, is the same object in both representations, so the cointegration content of VECM is fully accessible from PT-MV. Johansen's specification of the cointegration rank is replaced by the singular-value diagnostic of Definition 7.1.

### 8.5 Yule-Walker and the spectral methods

Yule-Walker and the related autocorrelation-function methods estimate $(\phi_1, \phi_2)$ by inverting the moment matching equations of a stationary AR($p$) process. They are inconsistent under non-stationarity (the autocorrelation function does not converge for I(1) processes), and the inconsis-

tency is severe enough to render them useless on the explosive and integrated regimes of Sections 6 and 7.4. The pure-time forward equation makes no stationarity assumption.

### 8.6 Robust and quantile estimators

Robust M-estimators (Huber), quantile regression, and Theil-Sen estimators address the orthogonal concern of *heavy-tailed residuals*, where OLS has unbounded influence. The graded action estimator does not by itself address that concern; in its level-1 form it inherits OLS's tail sensitivity. A natural combination is the robust pure-time estimator obtained by replacing $\sum(\hat{\varepsilon})^2$ with a Huber loss $\sum \rho_c(\hat{\varepsilon})$ on the level-1 residual, leaving the higher-grade machinery untouched. We do not develop this combination here; the connection is recorded as an open direction (Section 9).

### 8.7 Summary

The pure-time graded action estimator is not a competitor to OLS in stationary regimes (where it is algebraically equivalent), nor to robust estimators in heavy-tailed regimes (where it is orthogonal). Its substantive contribution is the unified treatment of the regime axis: stationary, integrated, explosive, mixed, cointegrated, all handled by a single OLS regression in the basis $(y_t, \Delta y_t)$ or $(\mathbf{y}_t, \Delta\mathbf{y}_t)$. The non-abelian extensions (graded loss, quadratic chaos, wedge projections) are accessible *additionally* when the embedding is full rank.

## 9 Open questions and programmatic position

### Theoretical

- A formal proof of the $\gamma$-Theorem (Theorem 3.1) via the Pascal matrix argument of the synthesis §5.2 plus a Hambly-Lyons uniqueness argument over the truncated Lyons signature.
- Quantification of the gain in Conjecture 5.1 in terms of geometric invariants of the attractor: topological entropy, Lyapunov dimension, Frenet-Serret curvature variance (the companion paper B).
- Extension to multivariate processes: the action functional generalises to $\mathrm{Cl}(3 \cdot d_y, 0)$ for $d_y$-dimensional series. The dimensional invariant becomes the rank of a $T \times 3d_y$ matrix.
- The variational principle on $\Theta$: under what regularity conditions do the first-order equations $\delta_\theta \langle V_\theta \rangle_g = 0$ define a symplectic structure on $T^*\Theta$, and what is the associated Hamilton-Jacobi flow?

### Implementation

- Whitening of the embedding: whether 'none', 'diagonal' ($\sigma_g^{-1}$), or 'full' Mahalanobis ($\Sigma_q^{-1/2}$) gives the best precision/optimisation trade-off. Open empirical question (`notes/00 §2.14, BL1`).
- Local optimiser: golden-section coordinate descent suffices for Battery D demonstrations but caps optimiser precision at $\sim 10^{-2}$ in some directions (e.g. phase $\phi$ in sine recovery). Newton's method with analytic Hessians, or a quasi-Newton scheme such as L-BFGS, would close this gap.

- Universal approximator backup: for models without closed-form $\delta_\theta V_\theta = 0$, training a Kolmogorov-Arnold network on the moment tensor of [Vázquez Broquá, MSDS] as universal inverse function. Open empirical work (`notes/00 §2.14, BL2`).

- Chaos-induced numerical instability: under chaotic dynamics (Hénon, ST4'), the higher grade losses become ill-conditioned at large $T$ due to exponential trajectory divergence. A shadowing-style regularisation (matching truncated short-horizon trajectories) is natural and adopted in the dynamical-systems identification literature.

### Programmatic

- This paper sits in the *estimation arm* of the Hamilton pure-time programme initiated in the synthesis. Together with Papers 28 (probability) and 29 (geometric invariants), it covers the three foundational faces of the programme: probabilistic (j-graded characteristic functions), geometric (Frenet-Serret / Egregium), and statistical (graded action).

- The OLS limit $w = (1, 0, 0, 0)$ is identified explicitly as the abelian, Cartesian projection of a richer multivectorial fit. The paper does not claim to *improve* OLS; it claims to *contain* OLS as a degenerate special case and to identify the precise structural condition (rank-3 embedding) under which the higher-grade content of pure time gives non-redundant identification information.

- The $\gamma$-Theorem provides an a-priori test, computable in one SVD pass, for whether a given dataset can benefit from graded-action estimation. This is the kind of operational invariant the GSD programme has been searching for.

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# A Post-submission addendum: rotor representation in Spin(3)

*This addendum reports an extension of the construction suggested by A. Sudjianto in personal correspondence following the initial submission of the manuscript. It does not alter any claim of the body of the paper; it offers, rather, a complementary representation of the central operator* $\hat{\Pi} = \hat{A} + \hat{B} - I$ *that resolves a numerical-stability weakness documented in §6.5 and that opens the path beyond the* $d = 3$ *ceiling that the Geometric Takens theorem of [Researches in GSD, Representation] imposes.*

### The construction

For any $\hat{\Pi} \in \mathbb{R}^{3\times 3}$ with singular value decomposition $\hat{\Pi} = U\,\Sigma\,V^\top$, the rotation matrix $UV^\top \in \mathrm{SO}(3)$ admits a rotor representation in $\mathrm{Spin}(3) \subset \mathrm{Cl}(3,0)$ via the standard half-angle construction. If $UV^\top$ has axis $\mathbf{n} = (n_1, n_2, n_3)$ and angle $\theta$, the corresponding rotor is

$$\mathcal{R} \;=\; \cos\tfrac{\theta}{2} \;+\; \sin\tfrac{\theta}{2}\left(n_3\, e_{12} - n_2\, e_{13} + n_1\, e_{23}\right) \;\in\; \mathrm{Spin}(3). \tag{10}$$

The rotor acts on a vector $\mathbf{v}$ by conjugation $\mathbf{v} \mapsto \mathcal{R}\,\mathbf{v}\,\widetilde{\mathcal{R}}$ and reproduces the original rotation matrix exactly. The decomposition of $\hat{\Pi}$ becomes the triple $(\mathcal{R}_L,\, \sigma,\, \mathcal{R}_R)$ with $\mathcal{R}_L, \mathcal{R}_R$ the left and right rotors and $\sigma \in \mathbb{R}^3_+$ the singular spectrum.

### Boundedness for free

The rotor norm satisfies $\|\mathcal{R}\|^2 \equiv 1$ identically by construction, since Spin(3) is a compact Lie group. Consequently, *any* divergence of the magnitude of $\hat{\Pi}$ is contained entirely in the singular spectrum $\sigma$; the rotation part lives bounded in Spin(3) regardless of the data scale.

We verify this property empirically. On a 3-dimensional panel with one explosive component (AR(1) with $\phi = 1.03$), one random walk, and one stationary AR(1), drawn at $T = 3000$, the explosive component reaches $|y_{1,T}| \approx 1.2 \times 10^{39}$ and $\hat{\Pi}$ attains condition number $\kappa(\hat{\Pi}) \approx 10^{77}$. Yet:

$$\|\mathcal{R}_L\| \;=\; 1.0000000000000000, \qquad \|\mathcal{R}_R\| \;=\; 0.9999999999999999.$$

The same property holds across forty orders of magnitude in input scale, from $10^{-20}$ to $10^{20}$. The directional content of $\hat{\Pi}$ is therefore representable as a stationary, bounded element of Spin(3), while the magnitude (which carries information about the cointegration spectrum) lives separately in $\sigma$.

### Implications

(i) **Resolution of the quadratic extension's blowup (§6.5).** The quadratic forward equation suffers numerical blowup under explosive extrapolation because the features $y_t^2$ scale as $\phi^{2t}$ and have no compactness mechanism. Re-expressing the rotational content of the estimator through the rotor lifts the construction into the compact group $\mathrm{Spin}(d)$, where blowup cannot occur. The factorisation $(\mathcal{R}_L,\, \sigma,\, \mathcal{R}_R)$ separates the bounded directional content from the unbounded magnitude, allowing each to be analysed under its own asymptotic regime.

(ii) **Path beyond $d = 3$.** The rotor construction (10) extends to $\mathrm{Spin}(d)$ for arbitrary $d$ via the exponential of bivectors: any rotation in $\mathrm{SO}(d)$ factorises into commuting plane rotations, each of which contributes its rotor partial as $\exp(\theta_i B_i/2)$ for a unitary bivector $B_i$ in the corresponding plane. The dimension of $\mathrm{Spin}(d)$ is $\binom{d}{2}$, quadratic in $d$, which is computationally manageable where

$\dim \mathrm{Cl}(d,0) = 2^d$ would not be. This construction is the bridge that A. Sudjianto identified in his bivector-component analysis programme [Sudjianto et al.(2026)].

(iii) **Implementation.** The rotor decomposition is implemented as a public primitive in the companion software repository under the identifier `svd_to_rotor`, with seven `pytest` units verifying machine-precision round-trip $\mathrm{SO}(3) \leftrightarrow \mathrm{Spin}(3)$, conjugation consistency, composition norm preservation, and boundedness across forty orders of magnitude.


## Acknowledgement

The author thanks A. Sudjianto for pointing out, in personal correspondence (May 2026), that the rotor representation in $\mathrm{Spin}(d)$ furnishes the natural compactness mechanism for the rotation part of $\hat{\Pi}$, and for sharing the operational construction beyond $d = 3$ that his bivector-component-analysis programme has developed.